CREDIT NIRVANA

Open IIT Data Analytics

PROBLEM STATEMENT 3

Address Geocoder

Team **PointWise**



PROBLEM STATEMENT AND ROADMAP

01

ADDRESS GEOCODER THAT LEARNS FROM FIELD VISITS

GOAL

Give every account address a map pin, a confidence radius, and landmark-based directions, while continuously improving location accuracy using multiple data sources, spatial patterns, address relationships, and real-world feedback.

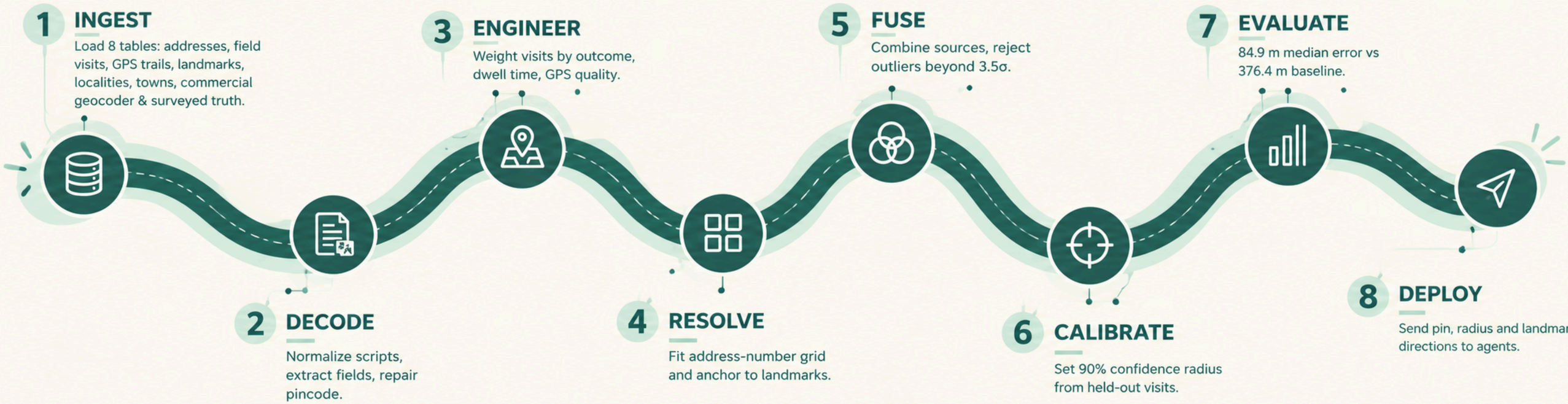
Key Challenges

- Visit GPS is a clue, not ground truth
- Mixed-script, misspelt Indian addresses
- Confidence scores must be honest
- Fake check-ins can mislead the model

Why It Matters

100s of m to km — error when commercial geocoders pin the pincode centre

Lost learning — visit GPS stays in logs and never updates the address



DATA UNDERSTANDING, EDA & DATA QUALITY

02

DATASET RELATIONSHIPS · MISSING & MALFORMED FIELDS · VISIT OUTCOMES · GPS TRAILS · EVIDENCE AVAILABILITY

3,117
addresses to geocode

2,880
commercial geocoder pins

100
surveyed ground-truth points

237
rural, out-of-town addresses

How the 8 CSV files connect

ADDRESS SIDE

- addresses
- localities
- towns
- landmarks

▼ linked by address_id

FIELD EVIDENCE

- field visits
- GPS points

▼ scored against

BENCHMARKS

- commercial pins
- surveyed truth

Visit outcomes = location reliability



Finding the building matters more than meeting the borrower, so every outcome except "not traceable" counts as location evidence.

GPS trails & data quality

2.5×
trail end is more accurate than the check-in point

150 s
dwell time from which a visit is reliable

Data quality issues we handle

- Mixed scripts, misspellings, abbreviations
- Malformed pincodes (e.g. 7 digits) repaired
- 3 address styles: cross/main, gali/ward, block/road
- Not every address has a visit: clusters and a grid fill the gaps

FEATURE ENGINEERING AND ADDRESS PARSING

03

Turning free-text addresses into structured information for accurate geocoding

Address Parsing Pipeline (parse_address)

Normalize text → Extract pincode → Extract landmarks → Extract grid details → Extract house number → Check rural markers → Resolve locality

Text Normalization (normalise)

1. Unicode NFC (standardize Indic text)
2. Lowercase (Latin only)
3. Replace # with space
4. Standardize house-number prefixes (h.no, house no, no. → houseno)
5. Remove punctuation and extra spaces

Landmark Extraction

Indic landmarks (SCRIPT_LANDMARKS)

आ

- ~30 Devanagari and ~25 Kannada entries mapped to canonical types

Examples
मंदिर, स्कूल,
बस स्टॉप, ಚರ್ಚ್

Latin landmarks (LATIN_LANDMARKS)

Aa

- Common place names (e.g. temple, school, water tank, bus stop)

Searched in normalized text

Grid Extraction by Town Style (extract_grid)

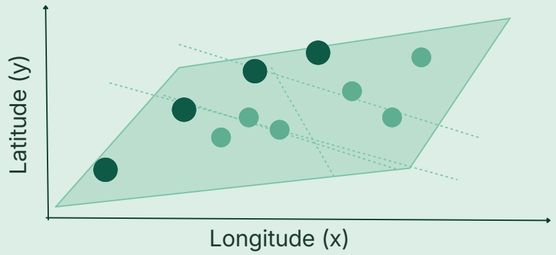
Town	Address Convention	Extracted Slots
T1 Kaveripura	Cross & Main streets	cross, main
T2 Devgarh Nagar	Gali & Ward	gali, ward
T3 Navanagara East	Block & Road	block, road

Parsed Output (ParsedAddress)

Field	Example Value
house_no	12
cross / main	6 / 5
locality_id	L101
landmark_types	['church']
has_proximity_cue	True
pincode	960102

Grid Model for Location Prediction

Learn an affine map from address numbers to coordinates
(using visit evidence and RANSAC for robustness).



Affine model (T1 example)

$x = a_1(\text{cross}) + a_2(\text{main}) + a_3$

$y = b_1(\text{cross}) + b_2(\text{main}) + b_3$

RANSAC Fitting (per town, locality)

- Need enough observations
- 120 trials, 90 inlier threshold
- At least 6 inliers
- Reject if median residual > 45 m

Parsing Performance

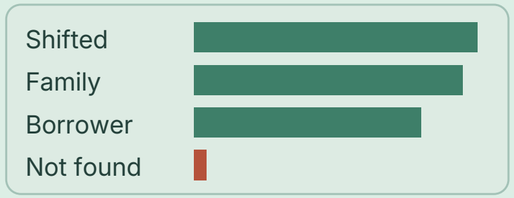
(in-town addresses)

Locality resolved **86.4%**	Pincode found **93.6%**
At least one landmark **59.5%**	Proximity cue **65.6%**

FIELD EVIDENCE ENGINEERING & ROBUST LOCATION ESTIMATION

NOISY VISIT GPS → TRUSTED EVIDENCE → ONE ROBUST LOCATION

Outcome reliability



- Credible outcomes weigh more
- Failed search = zero weight

×

Dwell time



- Longer stay, stronger signal
- Capped at 150 s

×

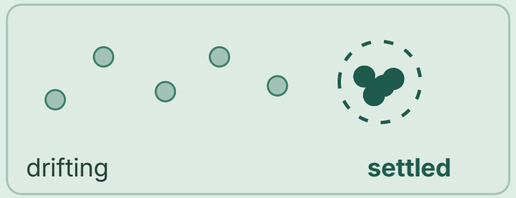
GPS accuracy



- Precise fix, higher weight
- Poor fix is downweighted

×

Trail settling



- Stable trail end is trusted
- Drifting end counts less

Combined visit weight

Outcome × Dwell

Accuracy × Settling

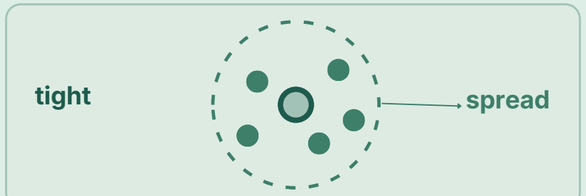
- One weak factor lowers it
- A score, not a probability

Geometric median



- Reliable visits pull harder
- Resists isolated outliers

Evidence dispersion



- How tightly the visits agree
- One visit → fallback value

OUTPUT PER ADDRESS

Robust coordinates
Evidence weight
Spatial dispersion

ENTITY RESOLUTION, EVIDENCE FUSION & OUTLIER DETECTION

MANY IMPERFECT SOURCES → WEIGHTED BY UNCERTAINTY → ONE SOURCE-AWARE LOCATION

Field visits

Very precise

- Weighted GPS visits from Slide 4
- Tightest when visits agree

Shared-address clusters

Medium

- Same locality + grid cell links records
- Own evidence excluded

Learned grids

Varies by town

- Grid numbers → coordinates
- RANSAC fit, only where data allows

Landmarks

Coarse

- Nearest matching landmark in locality
- Anchor when little else exists

Locality / Pincode

Coarsest

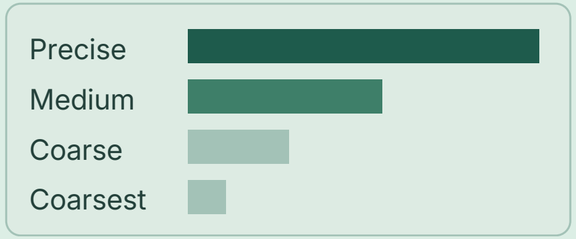
- Area centroids
- Fallback layer

Commercial geocoder

By pin type

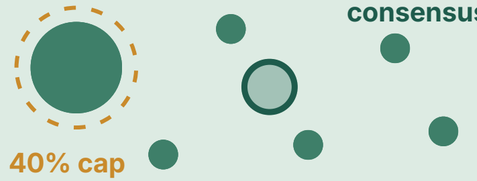
- Pins tagged by precision
- Street-level pins can be strong

Inverse-variance weighting



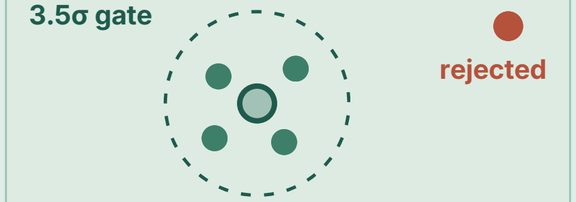
- Lower uncertainty → more influence
- Weights sources, not single visits

Robust consensus



- Robust centre of all sources
- Caps any one source at 40% of weight

Outlier gating



- Rejects sources far from consensus
- Needs ≥3 sources; keeps 2 tightest

FUSED LOCATION

Coordinates + source-aware uncertainty

Source disagreement widens it
→ 90% confidence radius

CONFIDENCE RADIUS & CALIBRATION

06

SOURCE UNCERTAINTY · HELD-OUT CALIBRATION · 90TH-PERCENTILE ERROR RATIO · FINAL UNCERTAINTY RADIUS

1 Source uncertainty

$$w = 1 / \sigma^2$$

Every source carries its own σ. Fusion gives one σ, widened when sources disagree.

2 Held-out calibration

$$q = \text{error} \div \sigma$$

5-fold: re-predict held-out addresses without their own visits, then compare.

3 90th-percentile ratio

$$k90 = Q0.90(q)$$

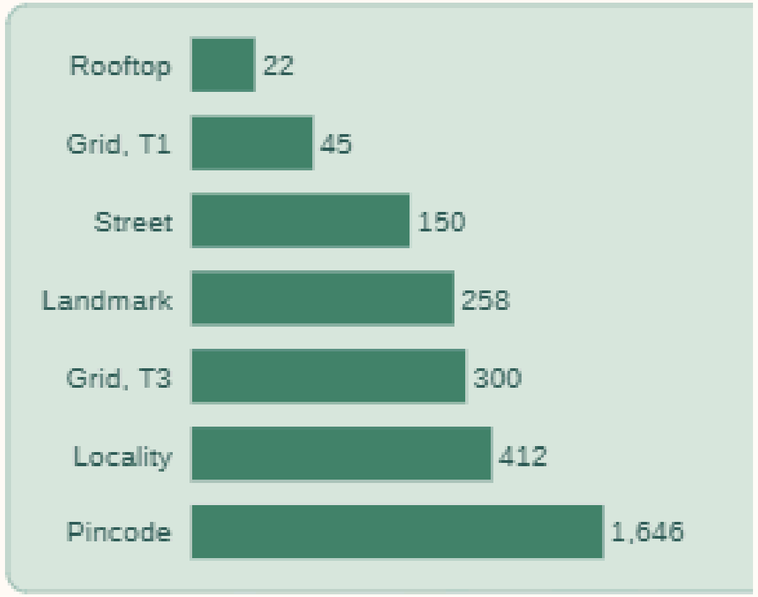
The ratio that 90% of held-out errors stay under.

4 Final radius

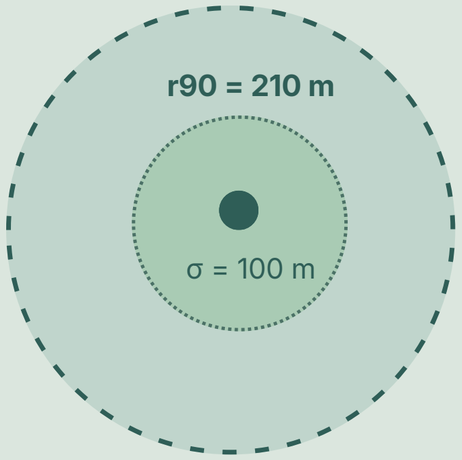
$$r90 = k90 \times \sigma$$

Clipped to stay between 20 m and 4,000 m.

Source uncertainty σ (m, log scale)



Worked example



100 m × 2.1 = 210 m

How the radius is used

Tight ring
Plan the visit directly

Wide ring
Verify the address first

PS2

Right-party contact
Radius feeds Problem Statement 2

Calibrated on held-out visit evidence; coverage then checked on the surveyed set.

EVALUATION, ABLATION, LEAKAGE CHECKS & LIMITATIONS

07

Commercial baseline comparison, error metrics, statistical tests, component removal experiments and honest limitations.

84.9 m

median error vs 376.4 m for the commercial geocoder (−77%)

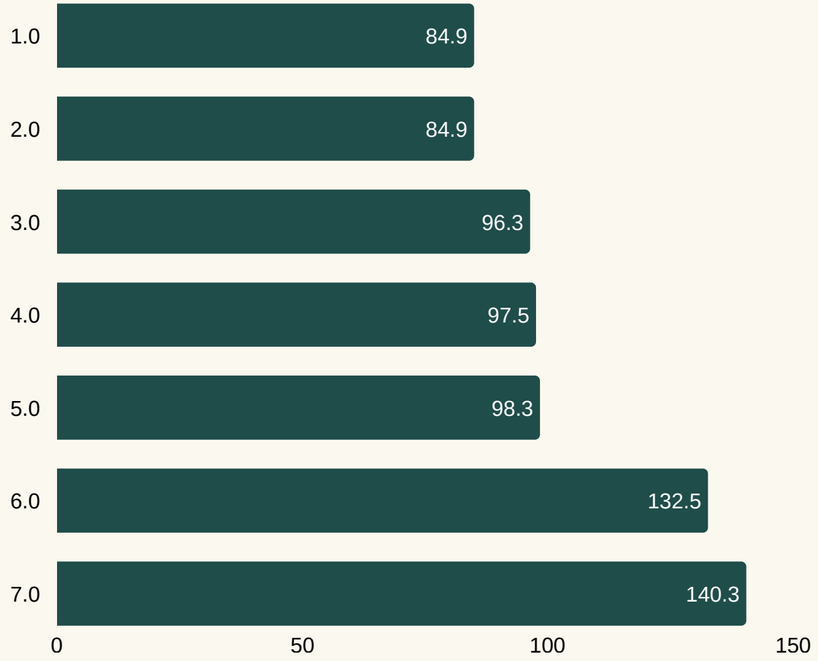
70%

of pins within 200 m vs 28% for the commercial geocoder

84 / 100

surveyed addresses where we are closer than the commercial pin

Ablation: median error (m) with one part removed



Biggest single gain: counting every visit that located the building, not only borrower-met ones.

Leakage check

We overwrite the surveyed coordinates with junk and the predictions do not change, so build() never reads them. EDA choices were tuned on surveyed data, so indirect leakage is not ruled out.

Limitations

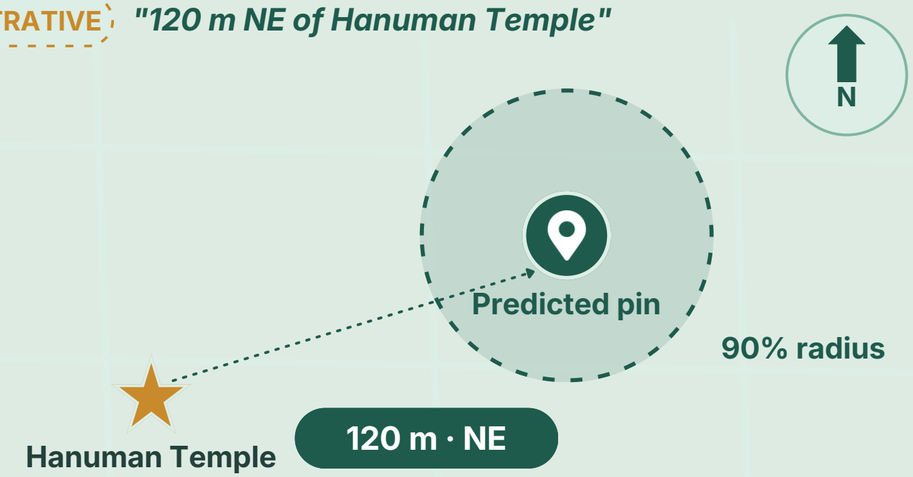
- **n = 100, not representative (median CI 57–136 m)**
- **Commercial-fallback tier wins only 44% of the time**
- **237 rural addresses flagged, not geocoded**
- **Synthetic data: numbers describe the generator too**

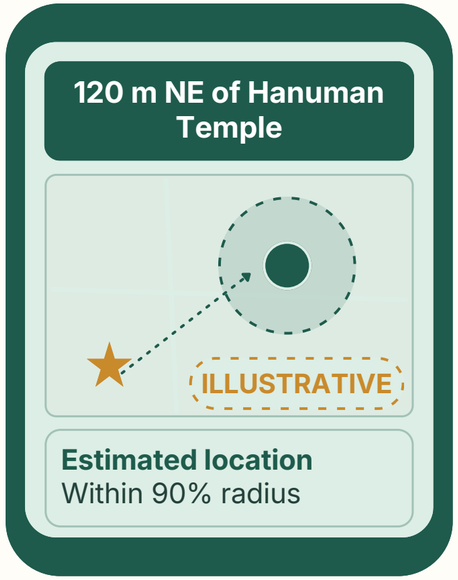
DEPLOYMENT, NAVIGATION & BUSINESS IMPACT

08

FROM PREDICTION TO ACTION IN THE FIELD

ILLUSTRATIVE *"120 m NE of Hanuman Temple"*





WHAT THE AGENT SEES

- **Predicted pin** — Model coordinates — IMPLEMENTED
- **Confidence radius** — Boundary of uncertainty — IMPLEMENTED
- **Landmark direction** — Distance + bearing — IMPLEMENTED
- **Offline field app** — Cached pin and directions — PLANNED

Field-agent workflow — PLANNED



Recorded outcomes sharpen tomorrow's pins (nightly retraining)

Platform integration — PLANNED

- **Visit planner** — Route visits on the pin
- **Low-confidence flag** — Verify before dispatch
- **Address records** — Write back confirmed locations
- **Right-party contact (PS2)** — Hard to find ≠ invalid

Benefits — EXPECTED

- Less time searching
- More efficient visits
- Better address data

Expected, not yet measured in production. Offline: 84.9 m median error vs 376.4 m commercial (100 surveyed, synthetic data).